

SURIYA PADITHAM

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PROFILE

Final-year Aerospace Engineering (Honours) student at UNSW with hands on experience across CFD, structural analysis and mechanical design, gained through a scramjet combustor mixing thesis and multidisciplinary team projects. Proficient in ANSYS CFX/Fluent, SolidWorks and MATLAB, with a strong grounding in aerodynamics and structural integrity. Seeking a 60-day industry internship to apply and extend this experience in a professional aerospace engineering environment.

EDUCATION

University of New South Wales <i>Bachelor of Engineering (Honours) (Aerospace Engineering)</i> Completed all required coursework and the thesis component of the degree. Requires a 60-day industry internship to satisfy final degree requirements.	<i>Kensington, NSW</i> <i>2022 – Present</i>
Sydney Boys High School — Higher School Certificate (Years 9 – 12)	<i>2018 – 2021</i>
Penrith Selective High School (Years 7 – 8)	<i>2016 – 2017</i>

TECHNICAL SKILLS

- **Software:** SolidWorks, Autodesk Fusion 360, ANSYS Workbench (Mechanical, Fluent & CFX), MATLAB, C Programming, Adobe Premiere Pro / Photoshop, Microsoft Office
- **Engineering Competencies:** Computational Fluid Dynamics, Aerodynamics, Structural Analysis, Mechanical Design, Technical Reporting

KEY PROJECTS

Investigating the Effect of Combustor Geometry on Mixing in Scramjet Engines — Honours Thesis

- Investigated the effect of passive combustor geometry (ramp and tab configurations) on fuel-air mixing behaviour in a scramjet combustor, using non-reacting CFD in ANSYS CFX benchmarked against the HIFIRE Flight 2 combustor.
- Modelled 10+ ramp and tab geometry variants in SolidWorks and simulated each under Mach 3 inlet conditions with the SST turbulence model on a 208,000+ element mesh.
- Identified that ramp geometries, particularly a concave profile, produced stronger and more effective mixing disturbance than tab configurations, informing design recommendations for future combustor development.

Motor-Actuated Passcode Safe — Team Design Project

- Designed the external enclosure in SolidWorks and ensured it was compatible with the electrical locking system.
- Collaborated within a team to integrate mechanical and electrical subsystems into a working prototype.

Wind Energy & Cyclone Risk Monitoring Project

- Co-developed a concept combining wind energy harvesting with real-time wind and weather monitoring for cyclone prone regions.
- Applied engineering design principles to balance structural resistance against extreme wind loading with energy generation performance.

Helicopter Winch Lug Structural Analysis & Manufacturing Project

- Designed and manufactured an aluminium helicopter winch lug using bolted and riveted joints, applying aerospace structural design practice.
- Conducted independent structural calculations under limit and ultimate load cases to validate strength and safety margins.

WORK EXPERIENCE

University of New South Wales — Open Day Student Assistant ● Engaged prospective students on engineering programs, coordinated event setup, and directed visitors throughout the venue.	<i>Kensington, NSW 2025</i>
Hindu Heritage Society — Volunteer ● Supported event setup, pack-down and venue preparation as part of a volunteer team delivering smooth event operations.	<i>Western Sydney, NSW 2015 – Present</i>

INTERESTS

Formula 1, Soccer, Cricket, 3D Printing, Gaming, Comics, Transformers, The Simpsons